



Unlocking Awareness:

How an Escape Experience Made Security Fun, Engaging, and Approachable



Joanna Behan

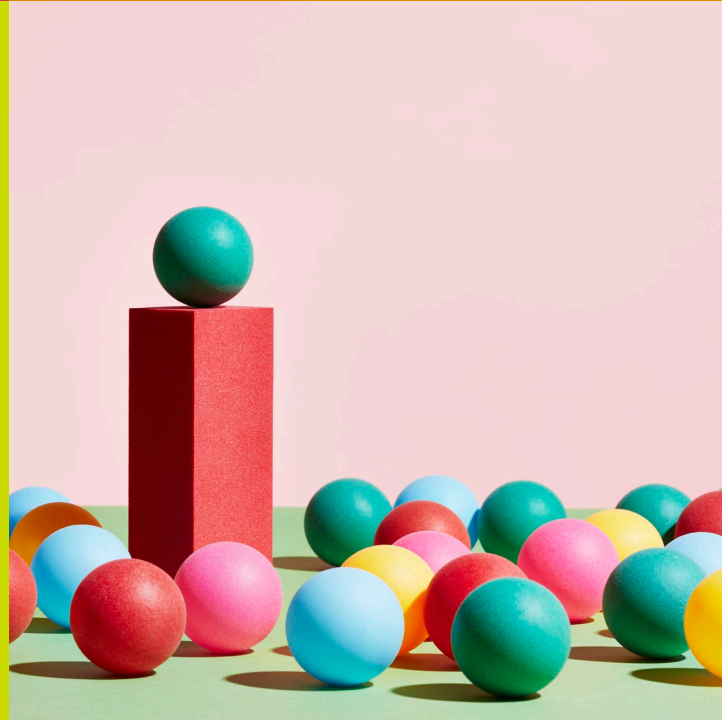
Agenda

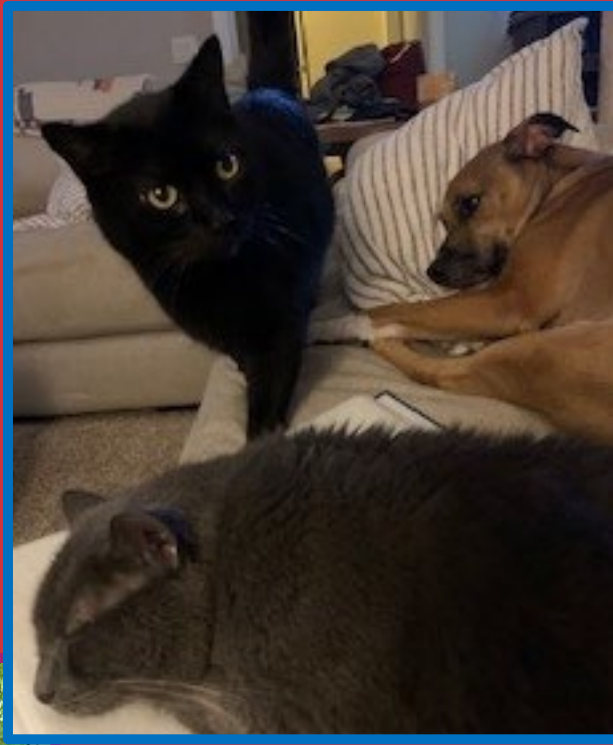
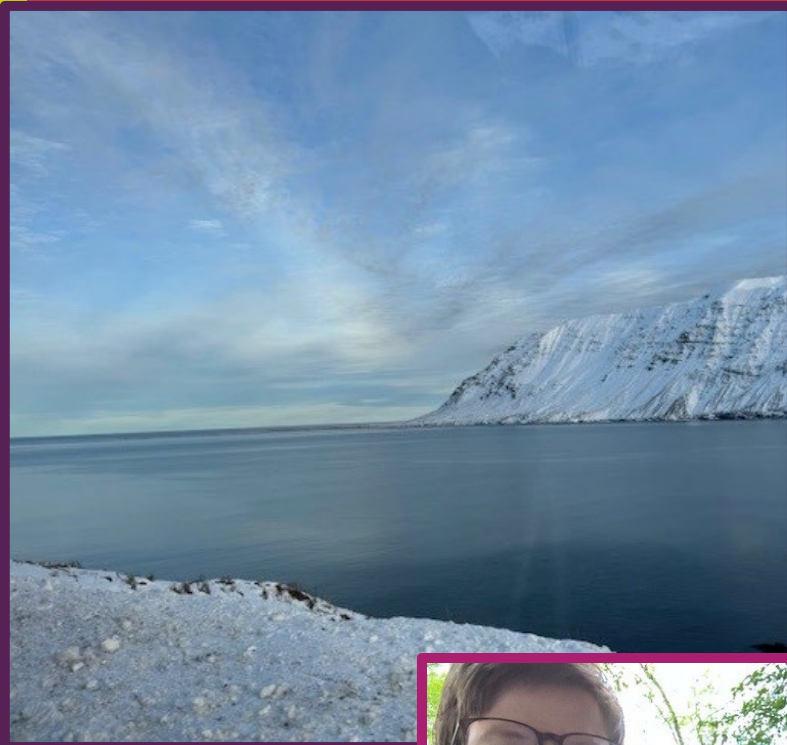
Introduction

Why a box?

Want to build your own?

Unlocked Wisdom





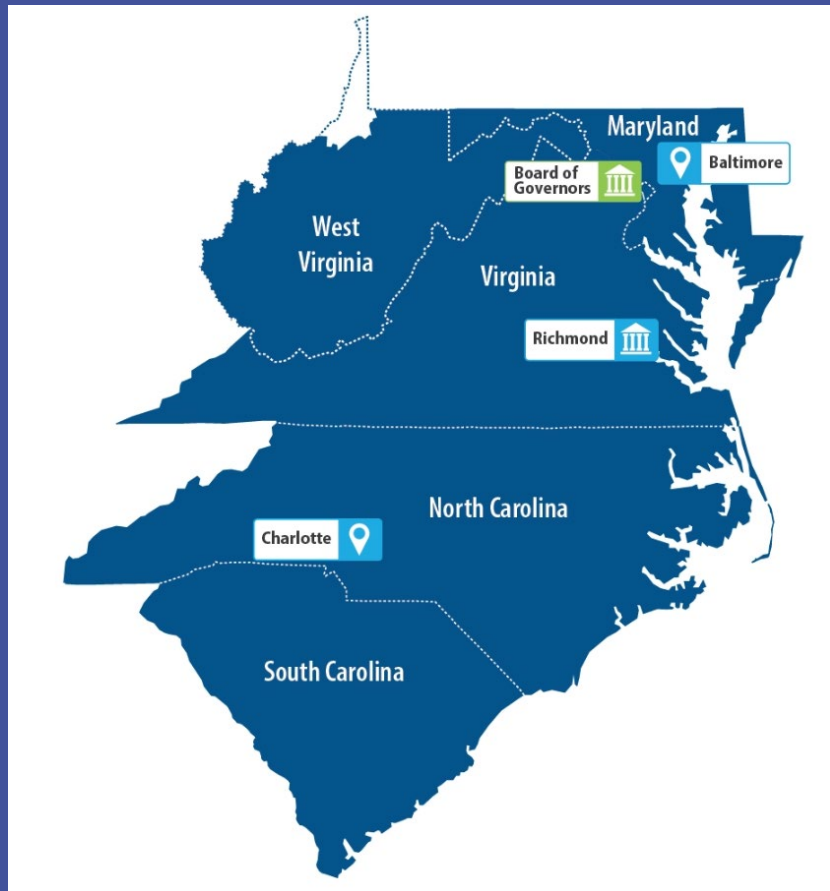
Hi, I'm Joanna Behan

The views expressed in this presentation are my own and do not necessarily represent the views of the Federal Reserve.



**Teamwork
makes the
Dream work!**

A bit about the **FRB of Richmond**

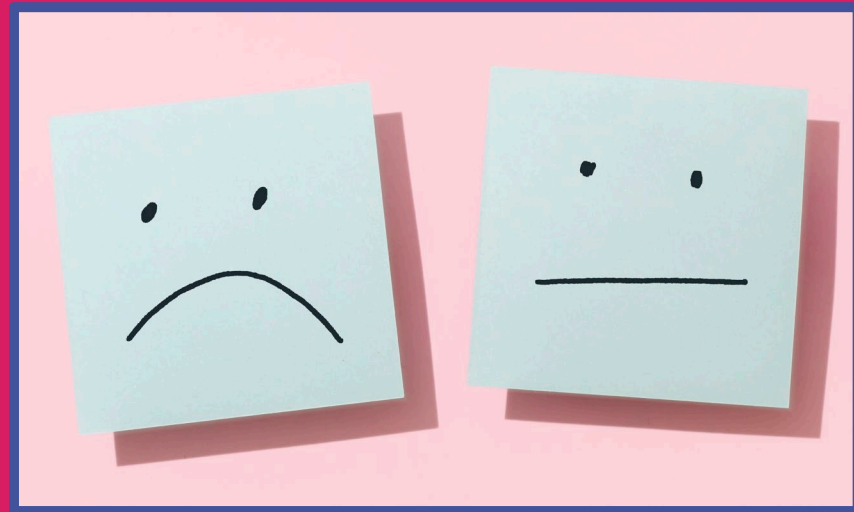


- [About Us](#) on [RichmondFed.org](https://richmondfed.org)
- [The Fed Explained](#) by the Board of Governors
- [Making Sense of the Federal Reserve](#) by the St. Louis Fed
- [About the Federal Reserve](#) on Federal Reserve Education
- [The Federal Reserve & You](#) by the Philadelphia Fed



Why a box?

Any of this sound familiar?



- Not another training...
- Uh oh, security is here to say no...
- Am I in trouble?
- I didn't know I couldn't do that...
- Do I have to?
- I don't have time for security...

Identified Goals & Opportunities



- **Target Cybersecurity Awareness Month (October 2025)**
- **We did not want to make light of cyber or information security; wanted a light & fun activity**
- **Information Security is a “team sport” & we are here to help**
- **No prior knowledge of topics necessary to participate**
- **Engage with Richmond & branches**
- **Think outside of the box**

BRIAN BURKE

GARTNER, INC

G → A → M

I → F → Y

HOW **GAMIFICATION**
MOTIVATES PEOPLE TO DO
EXTRAORDINARY THINGS

Why Gamify?

“Gamification is not just about applying technology to old engagement models...Gamification creates entirely new engagement models, targeting new communities of people and motivating them to achieve goals they may not even know they have.”

Brian Burke, *Gamify: How Gamification Motivates People to do Extraordinary Things* (p.4)

Thinking *inside* the box so you get to think *outside* the box.



- ✓ Mobile
- ✓ Scalable
- ✓ Reusable
- ✓ Customizable
- ✓ Affordable

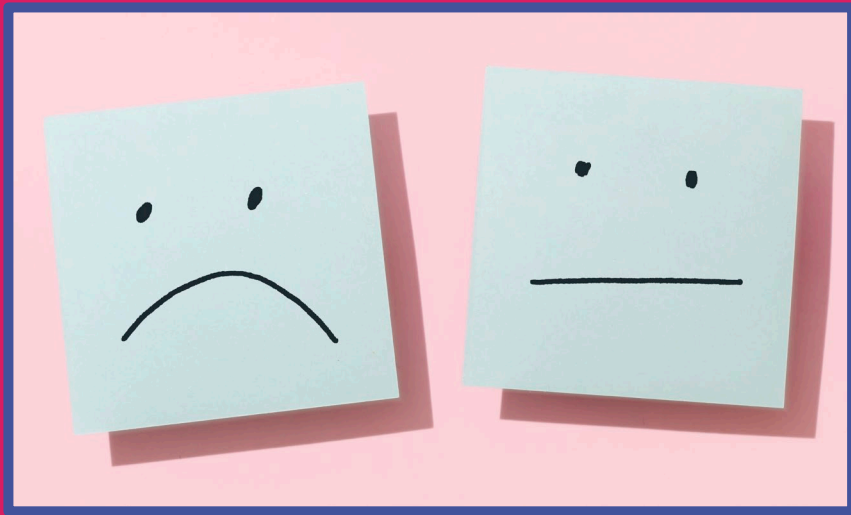
What **players** escaped with

- **Security can be fun**
- **Learned some new things**
- **Think outside the box (even in the box)**
- **Networking**
- **Communication and delegation**

What **Information Security** escaped with

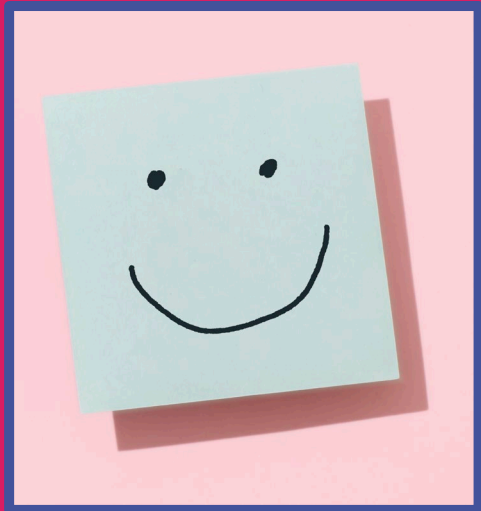
- **Security can be fun**
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- **Communication and delegation**

We changed this...

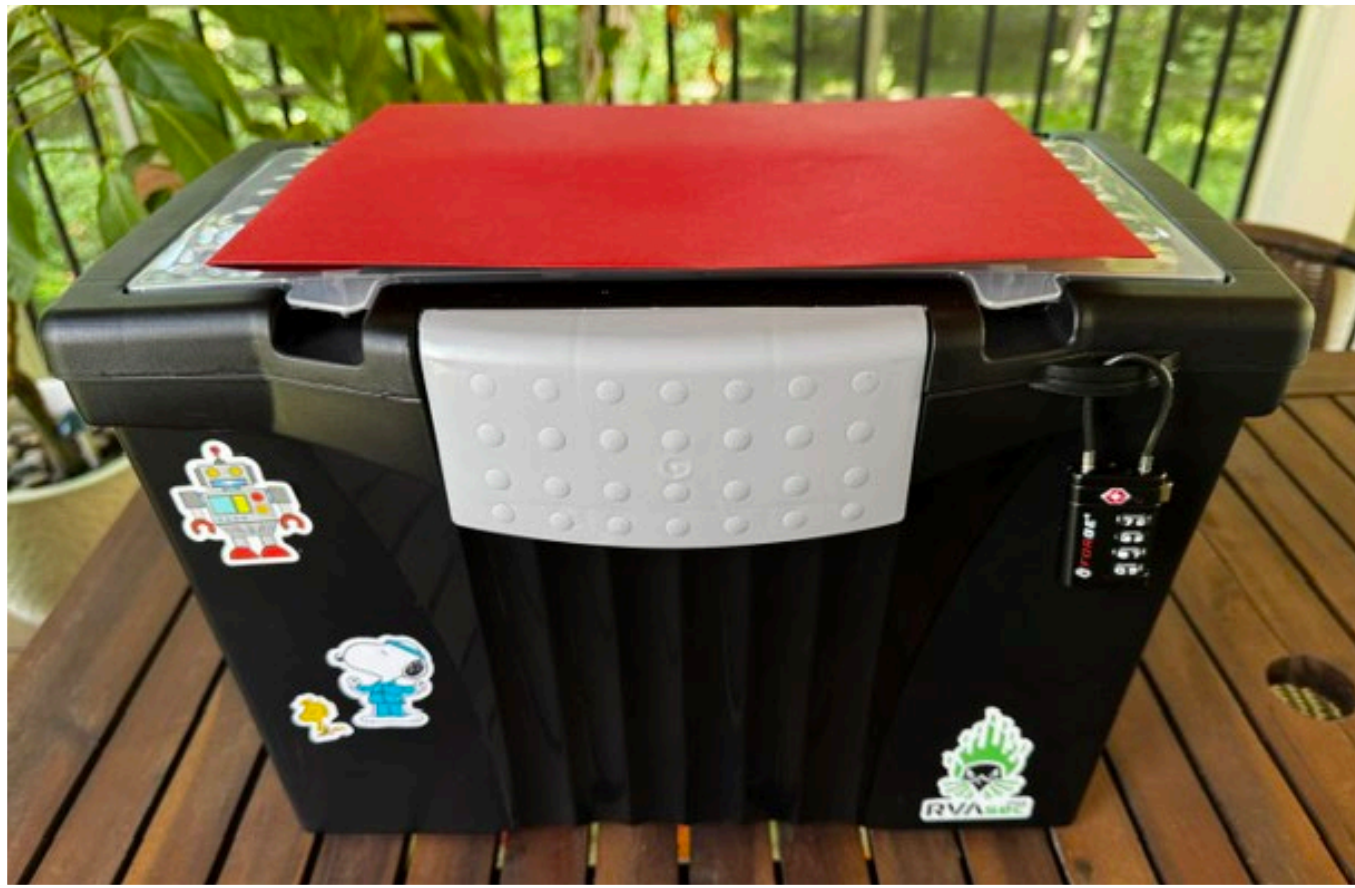


- Not another training...
- Uh oh, security is here to say no...
- Am I in trouble?
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- Do I have to?
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To this!



- ✓ I'm going to tell others to sign up!
- ✓ Do we have the new best time?
- ✓ I didn't know that before
- ✓ That was so much fun!
- ✓ It's over? I was hoping for more.
- ✓ Can I still play?
- ✓ Can I create my own?



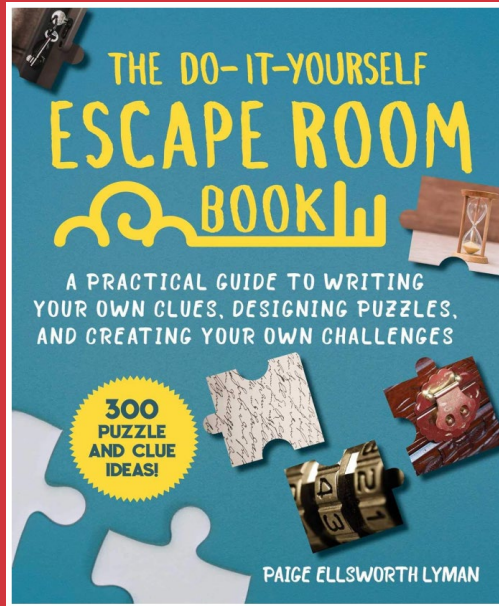
Ready to build your own box?

Step 1: Put things into a
box

Step 2: Set combination

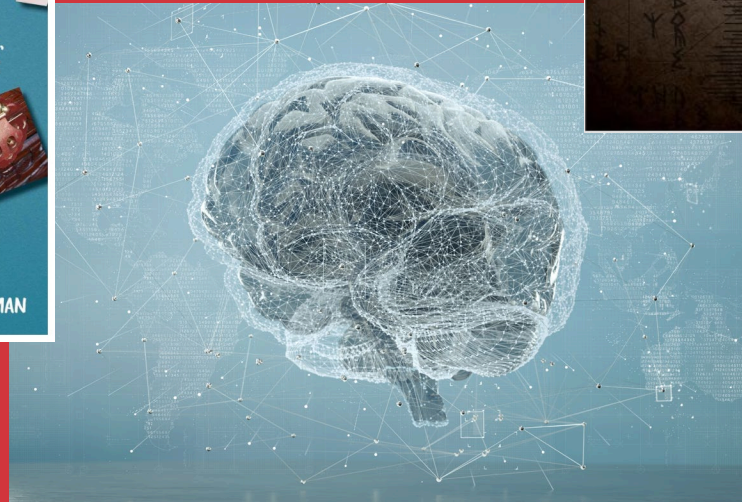
Step 3: Immediately
forget the combination

Research & Resources



Book

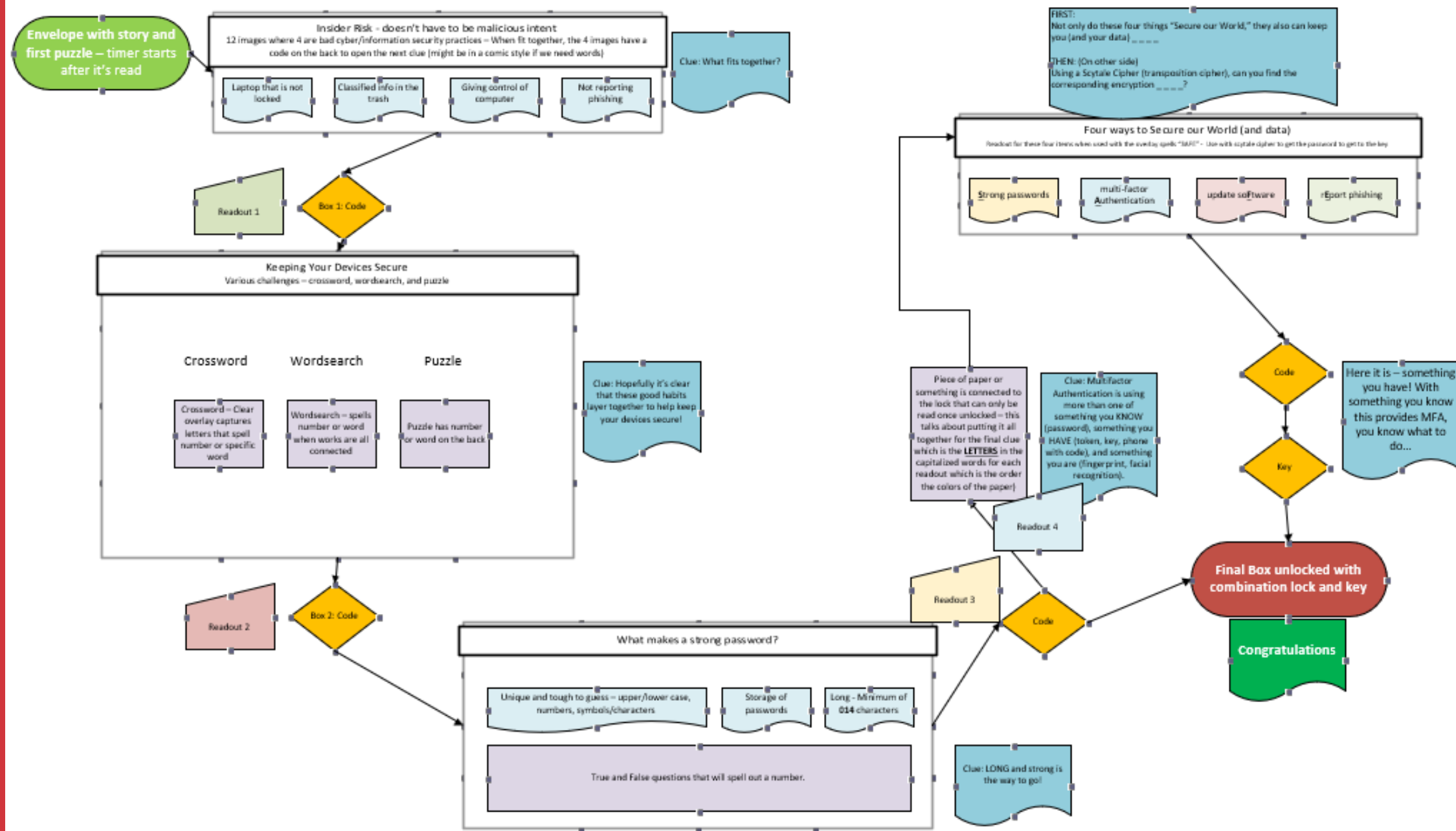
GenAI



Game

Theme





Supply List



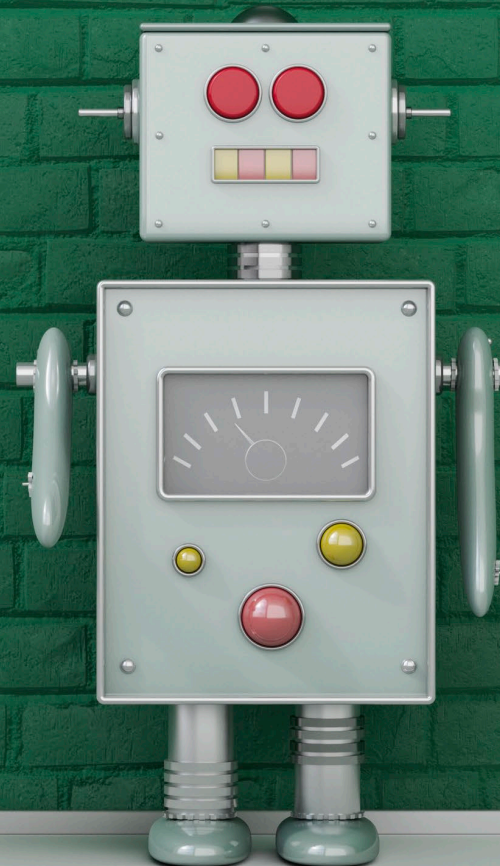
- Large lockable storage box
- Set-your-own combination lock (4-digit and 3-digit)
- Accordion file organizer
- Bag with set-your-own lock (3-digit)
- Small lockable container box
- Set-your-own word lock key lockbox (4-letter)
- Bag with key lock
- Puzzle (60 pieces or under)

- Expo markers
- Laminate sheets
- Paper (8x11 and 11x17)
- Ribbon
- Permanent marker or embroidery
- Dowel rod
- Envelopes (various sizes)
- Clear legal-size container or lid
- Stickers (because they are fun)



ALYSS and the Most Fun Ever Formula!

- Make sure to look at everything inside the box.
- Divide and conquer.
- Communicate effectively.
- Need a hint – ask your hosts!
- Don't share the solutions outside this room!



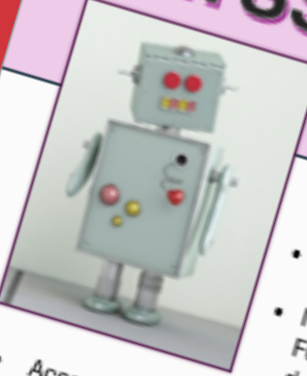
Let's
play!

The story

Envelope with story and
first puzzle – timer starts
after it's read



ALYSS and the Most Fun Ever Formula



- Stop! Don't go any further with this box until I can introduce myself!
- Hi – I'm ALYSS, which stands for, AI Likes Your Safety & Security – which I do!
- My primary programming is to keep this org's proprietary formula for the "Most Fun Ever" safe and secure. I have the autonomy to do what it takes to keep that data safe.
- According to the 2025 Verizon Data Breach Investigation Report¹, 60% of data breaches occur due to the human element such as credential abuse or susceptibility to social engineering like phishing.
- Other sources say that nearly 40% of individuals haven't changed their Wi-Fi password after they set it up, that '123456' is the most common password, and that 10% of people have used the same password since middle or high school – that can't be secure.
- Now I'm thinking YOU are a risk to this formula since you can access this data, even though in this scenario, it's your job to test this formula.
- With this logic, I've disabled your access! (It's how I was programmed)
- Can you identify the good (and bad) cybersecurity practices from four sets of puzzles and challenges? If so, I'll consider enabling your access again.

Once you are done reading this, your time will start, and you'll have 50 minutes to show me you know your stuff?

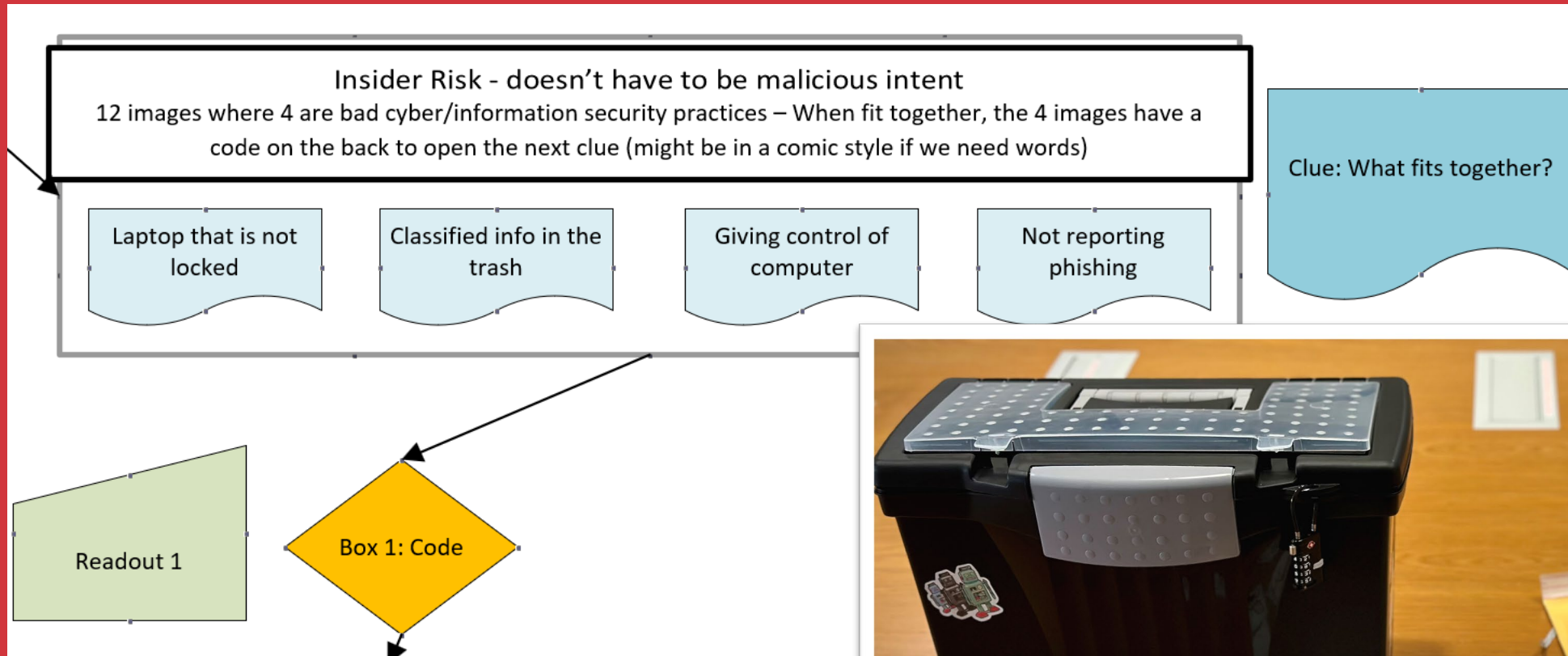
Hint for your first challenge – Can you identify the incorrect information security practices?

Good Luck! The timer starts now!

DBIR or <https://www.verizon.com/dbir>

This is a fictional story

Time starts now!



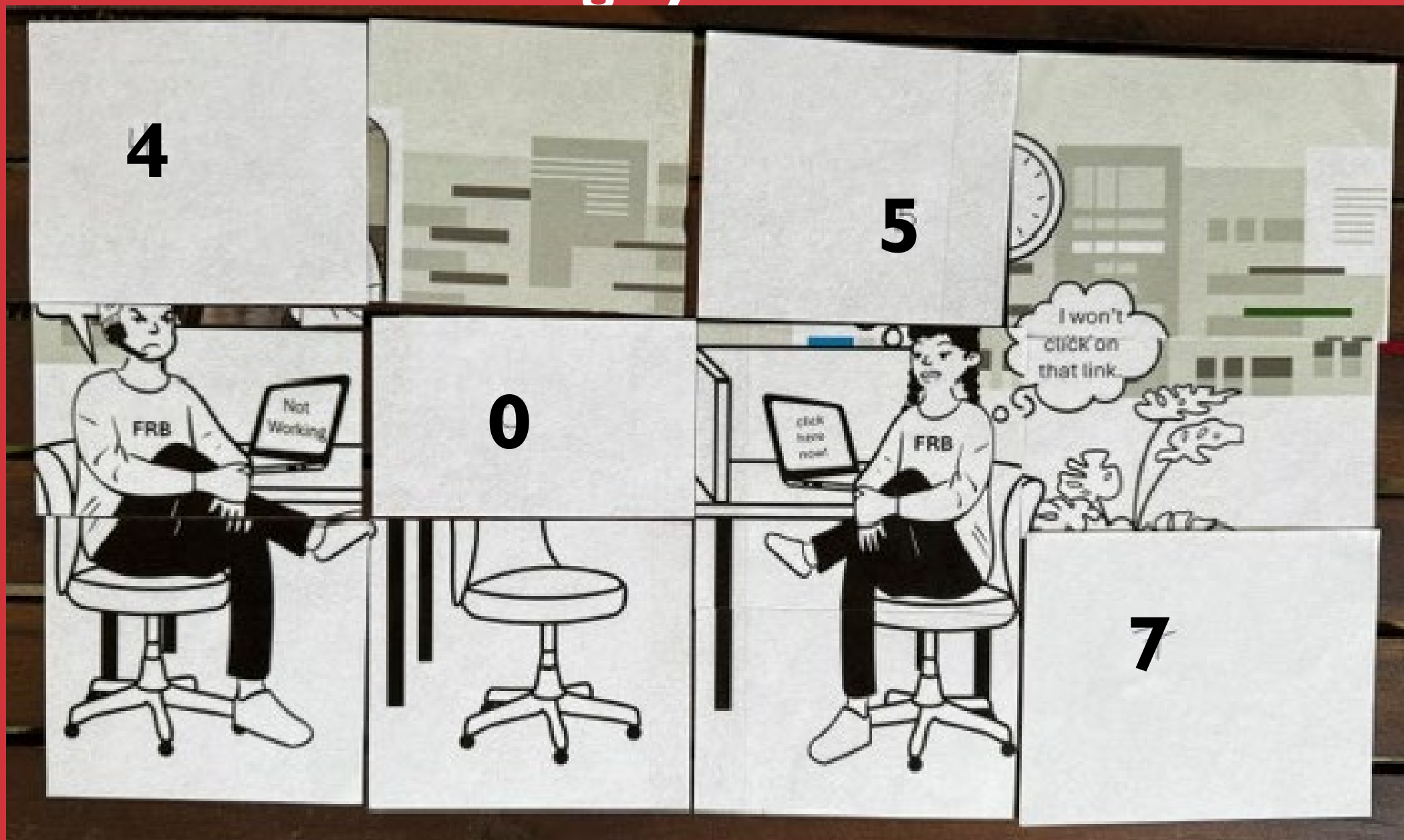
Lock 1: Which are things you shouldn't do?

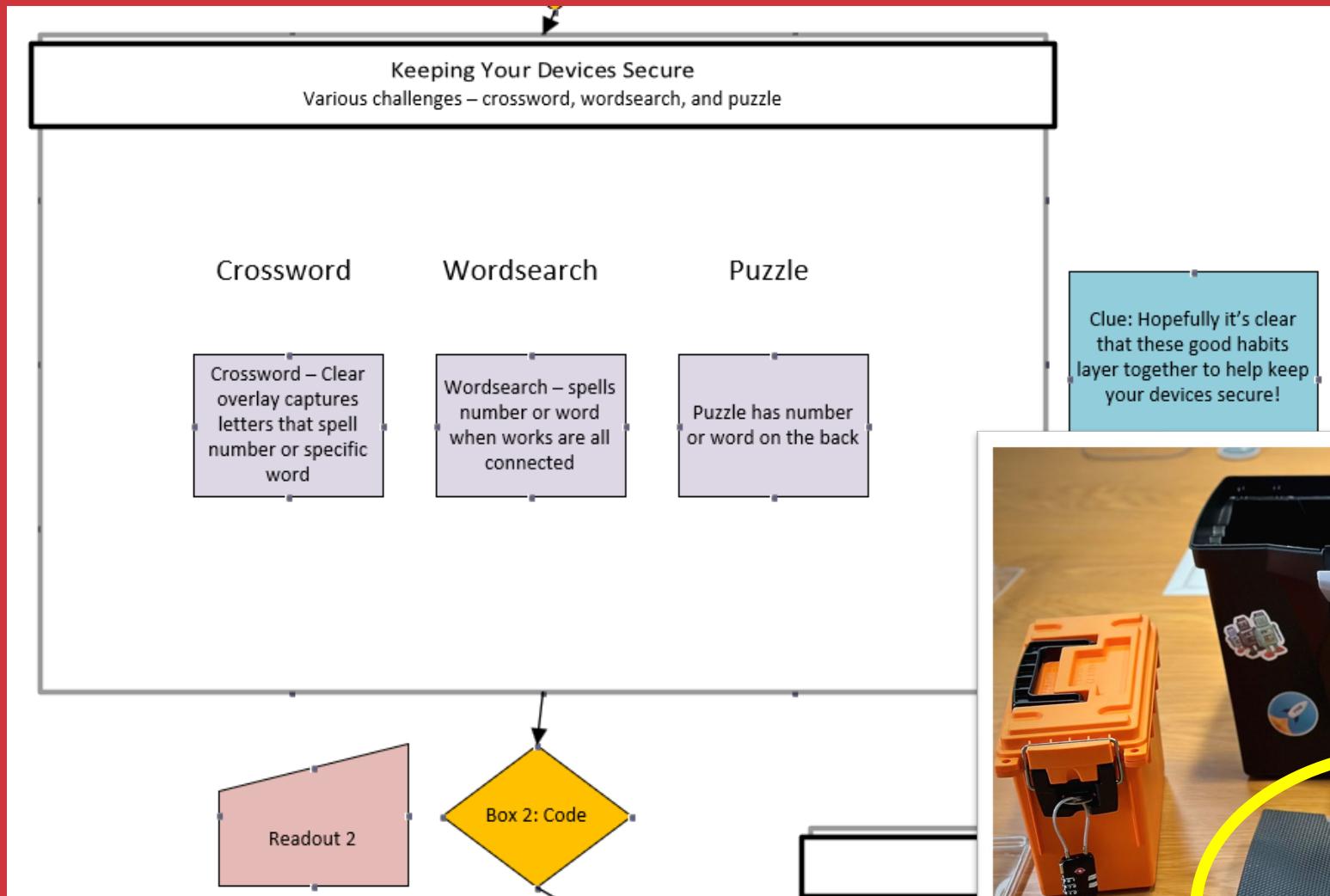


Lock 1: Which are things you shouldn't do? (created with Cartoon People in PowerPoint)



Lock 1: Which are things you shouldn't do?

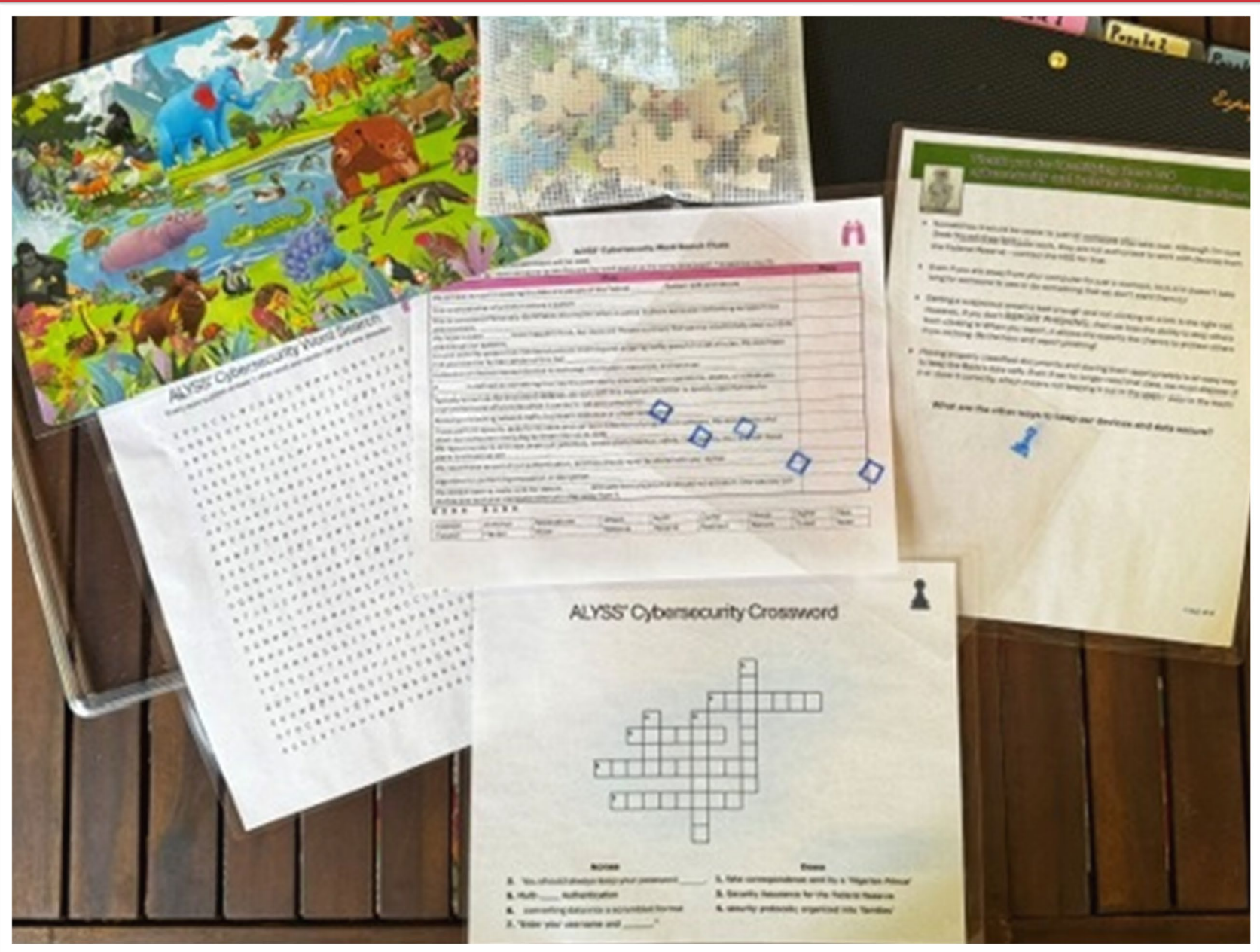




Lock 1 – Solved!



3 puzzles...



Puzzle 1

- Not all the words in the word bank will be used.
- Teamwork Tip – have someone tackle this and the word search at the same time (oops! I've said too much).

Clue	Word
We all have our part in keeping the data and people of the Federal _____ System safe and secure.	Reserve
This is where/what information enters a system. _____	Input
This is considered Personally Identifiable Information when it comes to yours but is also something we type in our web browsers. _____	Address
We hope a cyber _____ never happens to us, but there are threats out there that want to intentionally steal our data and disrupt our systems.	Attack
Crucial security system that monitors/controls incoming and outgoing traffic based on a set of rules. We also hope that you'll be the human version of this too! _____	Firewall
Collection of interconnected devices to exchange information, resources, and services. _____	Network
<u>A</u> _____ is defined as something that has the potential to adversely impact operations, assets, or individuals.	Threat
Typically known as the third line of defense, we work with this important function to identify opportunities for improvement and efficiencies when it comes to risk and compliance. _____	Audit
Allowing or blocking network traffic to prevent malicious or unwanted content. _____	Filter
These perform specific tasks for its users and can be a collection of programs or software. We also need to shut down our computers every <u>day</u> so these stay <u>up-to-date</u> . _____	Applications
We report incidents and risks (even just potential), create plans (backup, safety, contingency, etc.) and test those plans to ensure we are _____.	Resilient
We need these as part of our authentication, and they should never be stored with your laptop. _____	Token
Algorithm for performing encryption or decryption. _____	Cipher
We always want to make sure our data is _____ and safe from anyone that should not access it. One way you can do this is to lock your computer when you step away from it.	Secure

W O R D B A N K

Address	Antivirus	Applications	Attack	Audit	Cipher	Cloud	Digital	Filter
Firewall	Hacker	Input	Network	Reserve	Resilient	Secure	Threat	Token

Puzzle 1

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W O R D B A N K

Address	Antivirus	Applications	Attack	Audit	Cipher	Cloud	Dig
Firewall	Hacker	Input	Network	Reserve	Resilient	Secure	Thr

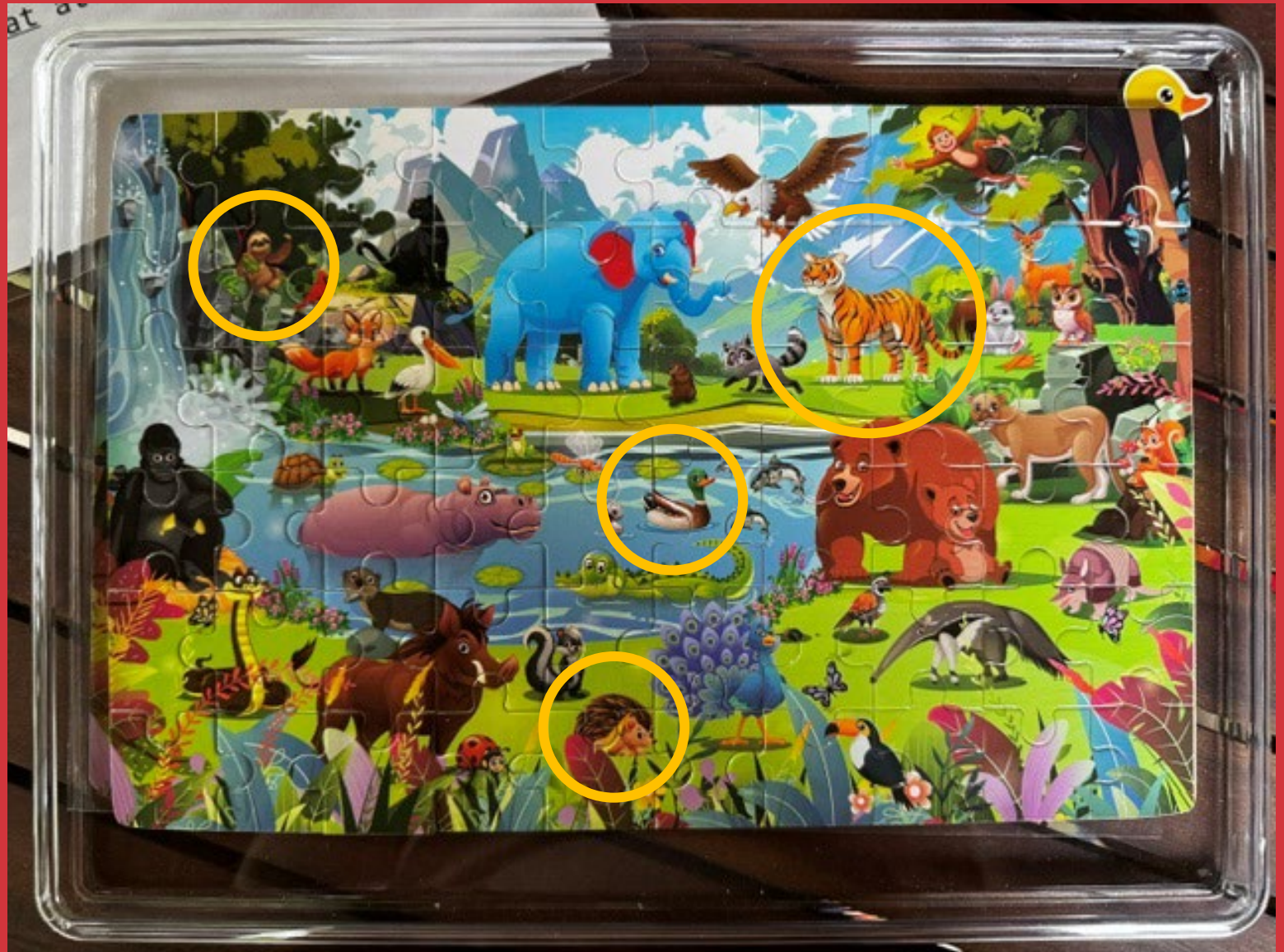
ALYSS’ Cybersecurity Word Search

Every word touches at least 1 other word and words can go in any direction.

E	P	U	C	C	X	L	W	Z	X	D	U	L	Q	B	M	U	X	Q	B	H	Y	P	J	D
U	N	B	R	E	H	P	I	C	Z	K	D	I	F	N	F	I	L	T	E	R	Z	Z	A	S
S	W	P	E	X	J	R	V	X	X	I	C	N	T	V	I	X	C	U	W	E	T	A	C	W
D	I	O	S	E	C	U	R	E	L	H	G	L	Q	A	R	Z	T	Y	C	S	L	U	N	C
E	X	R	E	P	E	V	I	J	U	L	M	F	U	B	E	F	Y	N	X	I	H	N	Z	H
K	K	K	R	G	K	P	Z	D	U	U	F	X	G	L	W	X	M	K	C	L	M	D	E	T
C	F	L	V	V	I	J	L	N	B	S	I	U	R	V	A	L	E	N	C	I	A	U	K	Q
A	O	T	E	A	R	D	J	E	Z	U	M	P	N	R	L	D	A	S	B	E	C	G	K	J
A	B	E	G	O	M	P	L	U	U	P	V	F	Y	V	L	Y	T	V	W	N	K	N	Z	Q
W	B	W	F	X	T	W	H	R	P	Y	U	C	R	T	H	R	E	A	T	B	M	S	C	
C	E	S	C	O	A	J	Y	B	C	R	U	M	Y	M	E	N	P	F	S	A	V	T	E	Z
J	G	G	N	H	A	L	A	R	M	E	E	Y	R	J	J	A	A	K	X	O	G	W	Q	P
U	Q	R	N	Y	Y	S	T	S	G	N	B	F	W	T	B	Q	P	U	I	E	Y	J	E	A
K	T	J	E	J	C	P	R	S	E	J	S	B	K	P	X	K	P	V	D	N	D	J	X	P
P	N	E	T	Y	L	Y	P	E	C	H	Y	M	I	P	T	C	L	Q	L	I	B	K	C	S
O	Q	N	W	M	T	Y	F	R	M	U	A	Q	S	P	A	I	I	M	X	I	T	B	J	Q
P	B	K	O	A	A	R	N	D	K	Q	P	F	P	Q	N	C	C	L	V	U	Y	J	O	J
S	R	A	R	M	V	M	S	D	B	I	F	W	T	F	D	P	A	B	P	E	G	D	C	G
Y	V	T	K	C	A	T	T	A	Z	O	K	F	J	F	Z	J	T	N	G	K	Q	U	L	A
N	U	O	T	M	Q	D	Q	V	U	O	P	J	V	O	Y	A	I	O	Y	G	K	P	Y	P
E	E	V	M	B	W	D	E	Q	Q	X	T	I	I	E	F	B	O	N	K	G	V	D	B	W
H	F	O	C	W	P	K	Y	S	N	B	O	X	W	B	G	Q	N	C	W	E	A	M	Y	N
G	G	V	I	K	T	Y	N	H	Y	B	M	G	Y	D	H	O	S	R	L	E	N	I	V	G

Puzzle 2

Cybersecurity can be a prickly business! You'll have to practice and pay attention to earn your stripes. Remember to take things slow so you don't make careless mistakes. When you master these techniques, you'll be able to stay afloat!

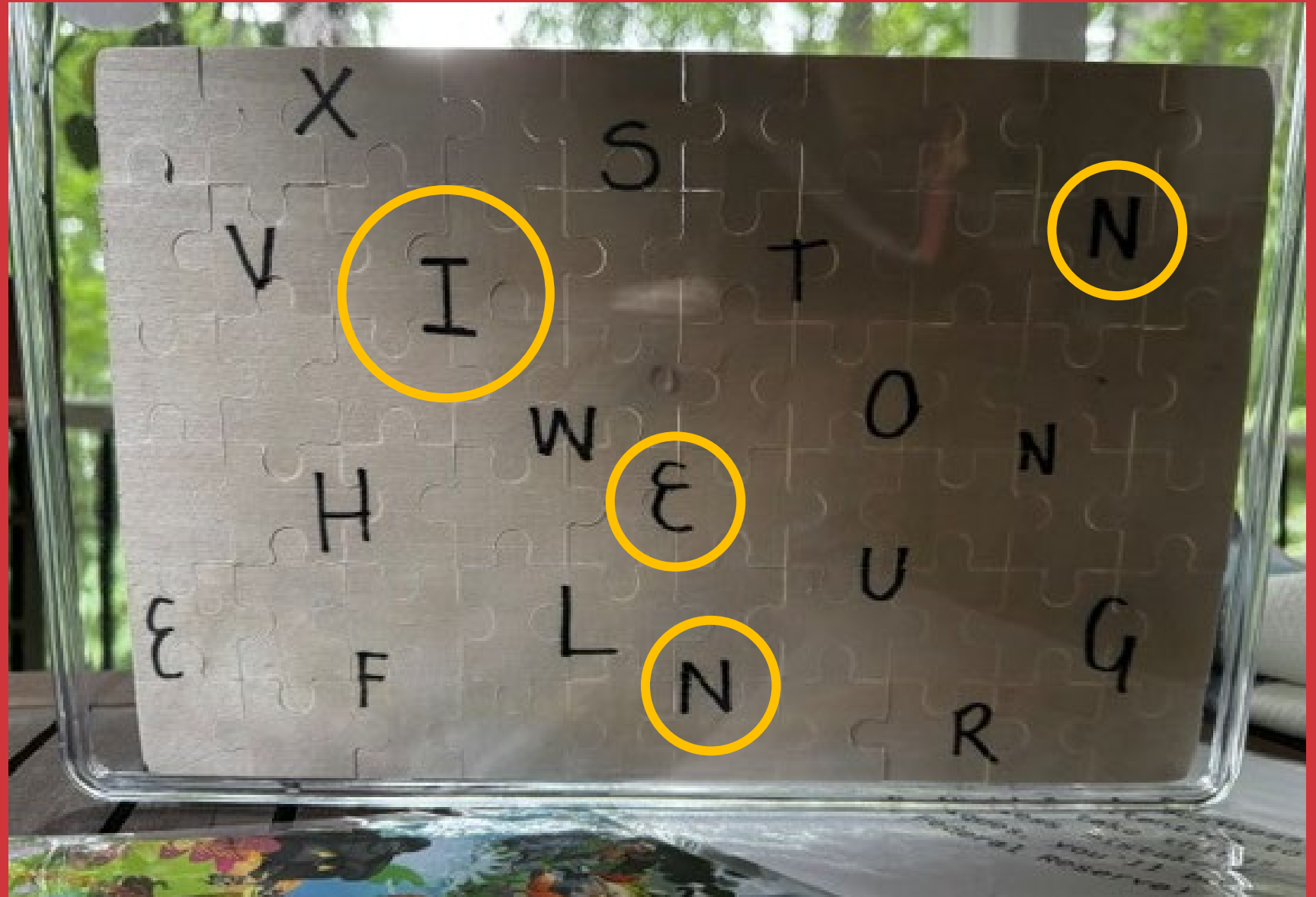


Puzzle 2

Cybersecurity can be a **prickly** business!

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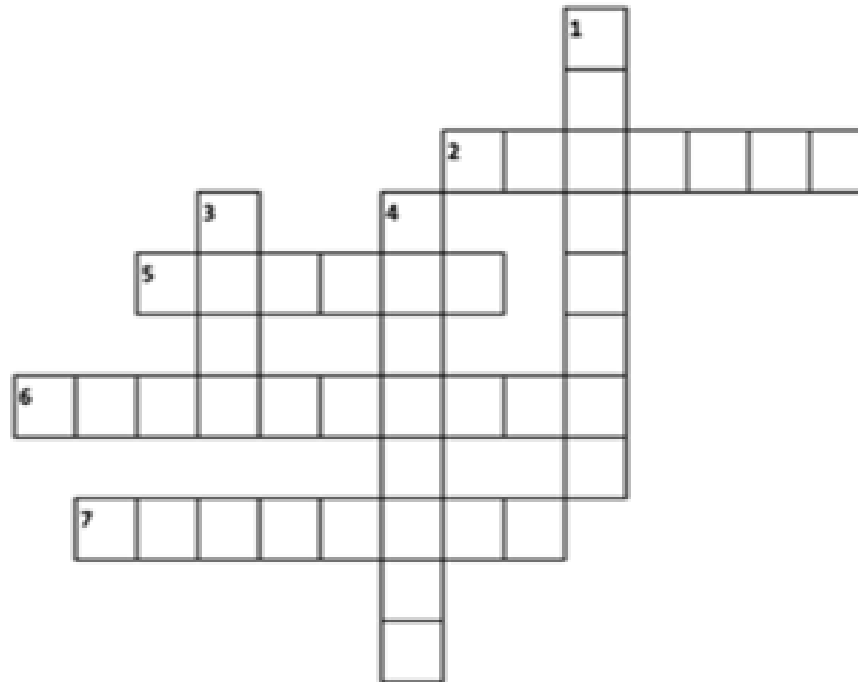


Puzzle 3

NONCONFIDENTIAL // EXTERNAL



ALYSS' Cybersecurity Crossword



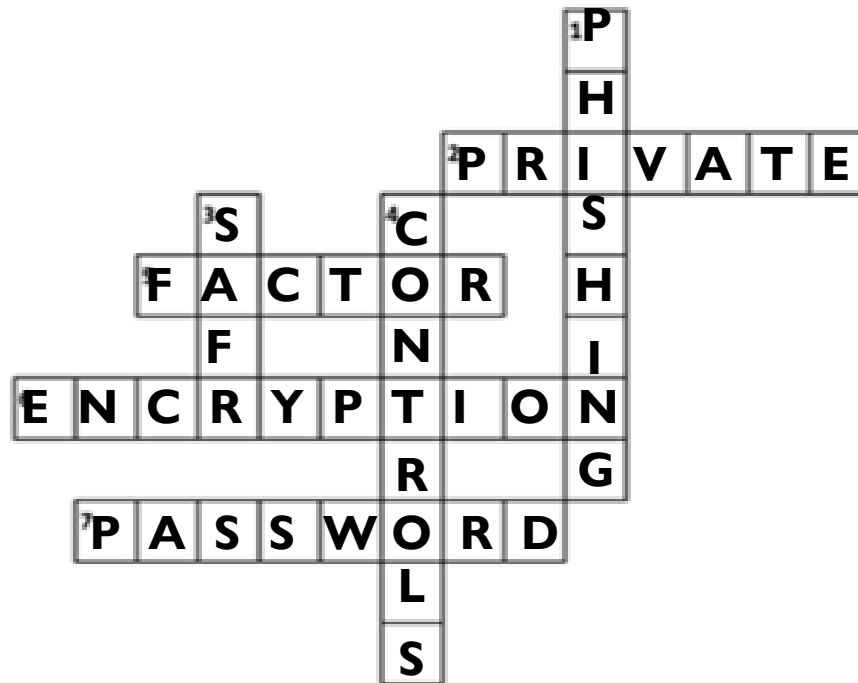
Across

- 2. You should always keep your password _____.
- 5. Multi-_____ Authentication
- 6. _____ converting data into a scrambled format
- 7. "Enter your username and _____."

Down

- 1. fake correspondence sent by a 'Nigerian Prince'
- 3. Security Assurance for the Federal Reserve
- 4. security protocols; organized into 'families'

ALYSS' Cybersecurity Crossword



Down

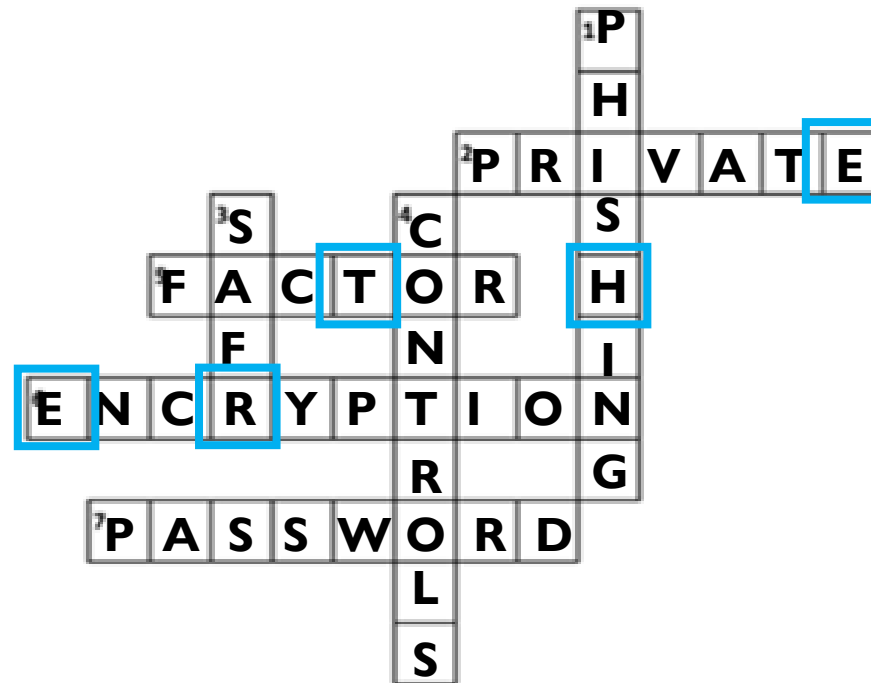
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Puzzle 3

NONCONFIDENTIAL // EXTERNAL



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Across

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Puzzle 3

ALYSS' Cybersecurity Word Search

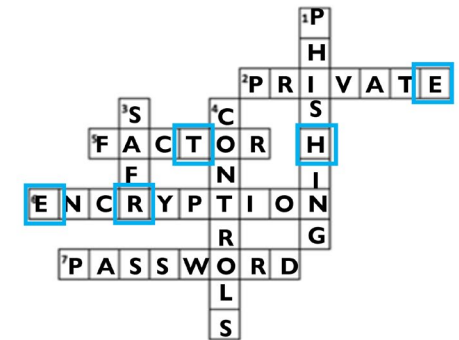
Every word touches at least 1 other word and words can go in any direction.

E P U C C X L W Z X D U L Q B M U X Q B H Y P J D
 U N B R E H P I C Z K D I F N F I L T E R Z Z A S
 S W P E X J R V X X I C N T V I X C U W E T A C W
 D I O S E C U R E L H G L Q A R Z T Y C S L U N C
 E X R E P E V I J U L M F U B E F Y N X I H N Z H
 K K K R G K P Z D U U F X G L W X M K C L M D E T
 C F L V V I J L N B S I U R V A L E N C I A U K Q
 A O T E A R D J E Z U M P N R L D A S B E C G K J
 A B E G O M P L U U P V F Y V L Y T V W N K N Z Q
 W B W F X T W W H R P Y U C R T H R E A T B M
 C E S C O A J Y B C R U M Y M E N P F S A V T
 J G G N H A L A R M E E Y R J J A A K X O G W
 U Q R N Y Y S T S G N B F W T B Q P U I E Y J E A
 K T J E J C P R S E J S B K P X K P V D N D J X P
 P N E T Y L Y P E C H Y M I P T C L Q L I B K C S
 O Q N W M T Y F R M U A Q S P A I I M X I T B J Q
 P B K O A A R N D K Q P F P Q N C C L V U Y J O J
 S R A R M V M S D B I F W T F D P A B P E G D C G
 Y V T K C A T T A Z O K F J F Z J T N G K Q U L A
 N U O T M Q D Q V U O P J V O Y A I O Y G K P Y P
 E E V M B W D E Q Q X T I I E F B O N K G V D B W
 H F O C W P K Y S N B O X W B G Q N C W E A M Y N
 G G V I K T Y N H Y B M G Y D H O S R L E N I V G



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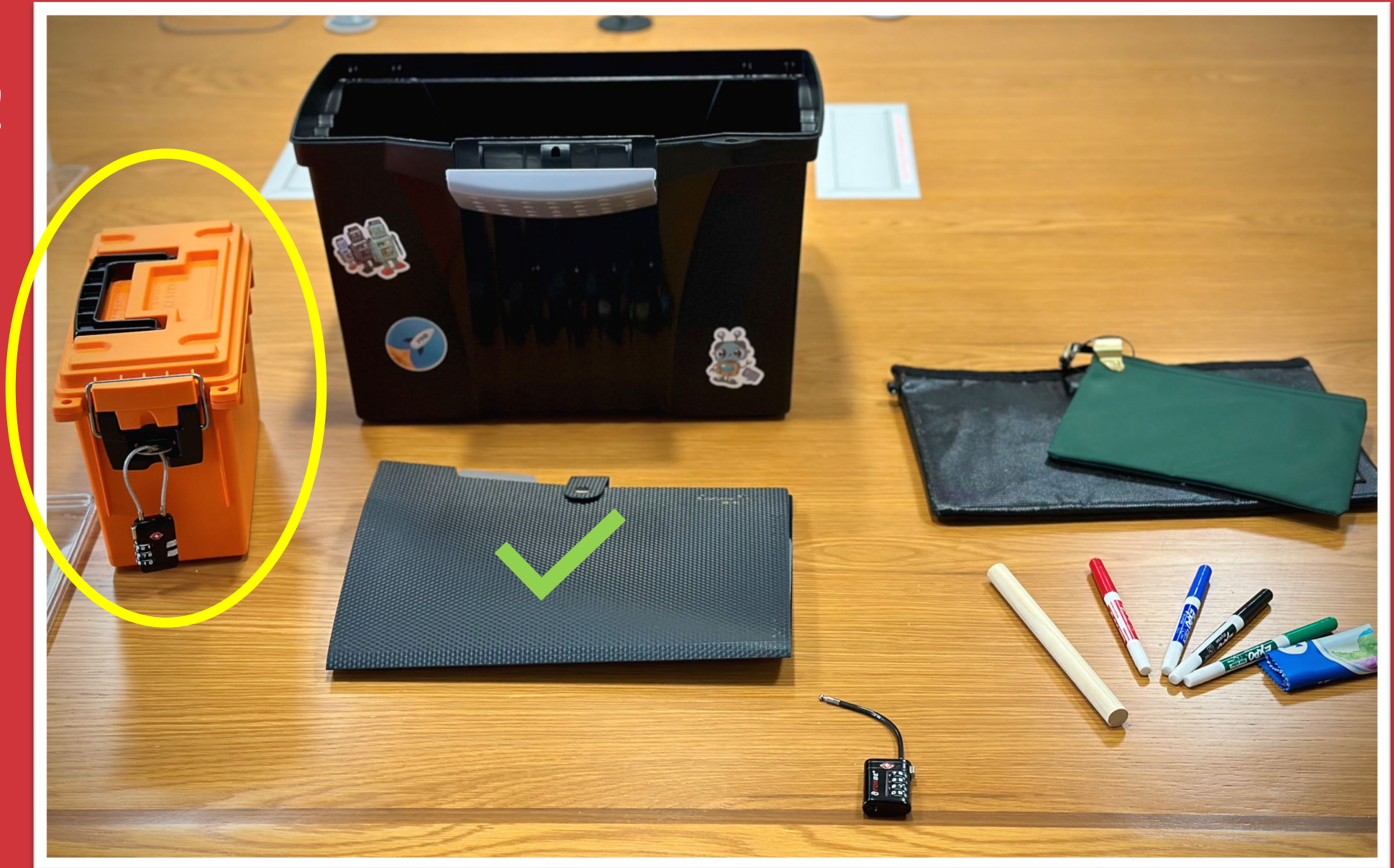
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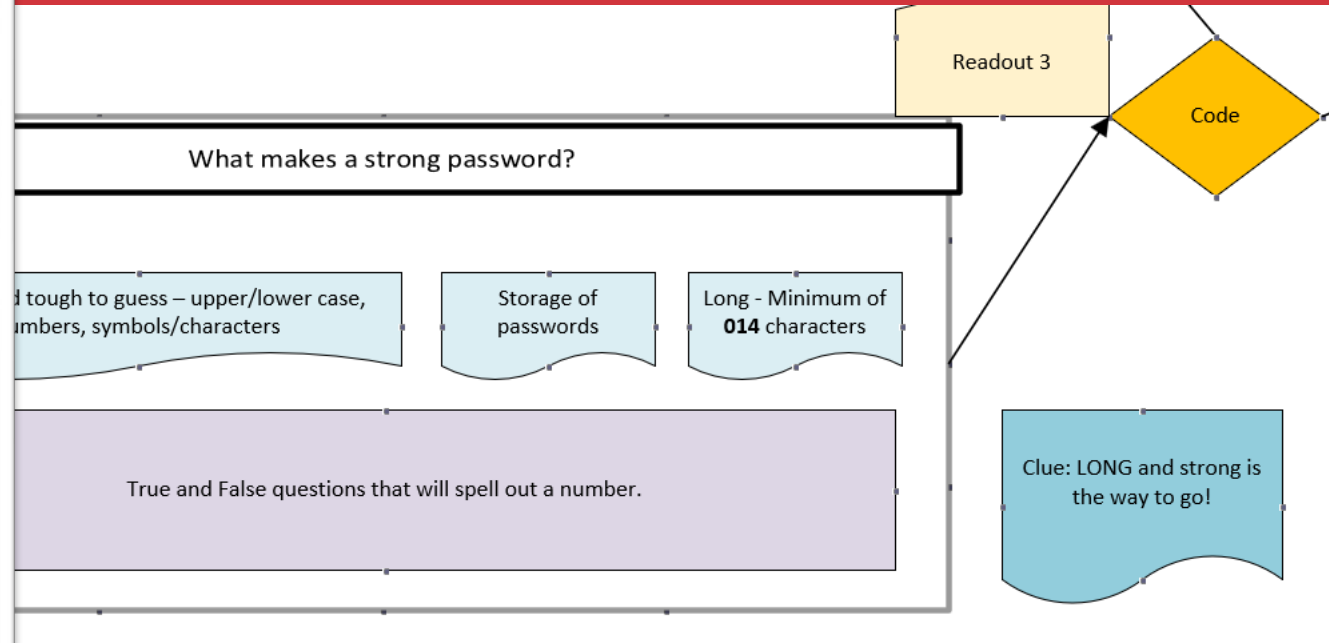
Down

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Lock 2 Solved!



All about passwords



A1: Every password should be unique.	D1: You should use the same password for everything.	J1: 10% of people have used the same password since middle or high school.
A2: If your password was part of a breach, then that password will always be vulnerable.	D2: Sharing a password is fine if it's with someone I trust.	J2: Credential abuse (to include passwords) is the most common vector for breaches.
A3: Enabling multifactor authentication is better than just a password.	D3: When you must change a password, changing just one character is good enough.	J3: Your birthday is a good thing to put in your password.
B1: Scripts should never contain sensitive credentials (secrets, tokens, etc.).	E1: Passwords should not be easy to guess and mix it up! For example, 1 for 'l' or @ for 'a'	K1: I can have a file on my work computer with passwords if it's password protected.
B2: If I keep the notebook with me all the time, it's ok to write down my passwords.	E2: Passwords should have numbers.	K2: Passwords need to be changed regularly.
B3: 60% of data breaches occur due to the human element which includes credential abuse.	E3: Since 2013, the most common password is '123456'	K3: A good password will have your pet's name in it.
C1: Nearly 40% of individuals haven't changed their Wi-Fi password after they set it up.	F1: Once you come up with a good password, you never need to change it.	L1: Saved plain text passwords in my OneDrive and/or OneNote are not protected.
C2: Passwords should have upper and lowercase letters.	F2: Plain dictionary words are great for a strong password.	L2: Passwords should be a combination of upper & lowercase letters, & special characters/symbols.
C3: A password manager can help manage passwords.	F3: Keyboard patterns like 'asdfgh' are good for passwords since it's not a real word.	L3: Passwords should be a minimum of 14 characters.

True or False?

Three columns means three people can review true/false statements at the same time

**If it's true, color it though.
If it's not, don't give it a spot.**

	A	B	C	D	E	F	J	K	L
1									
2									
3									



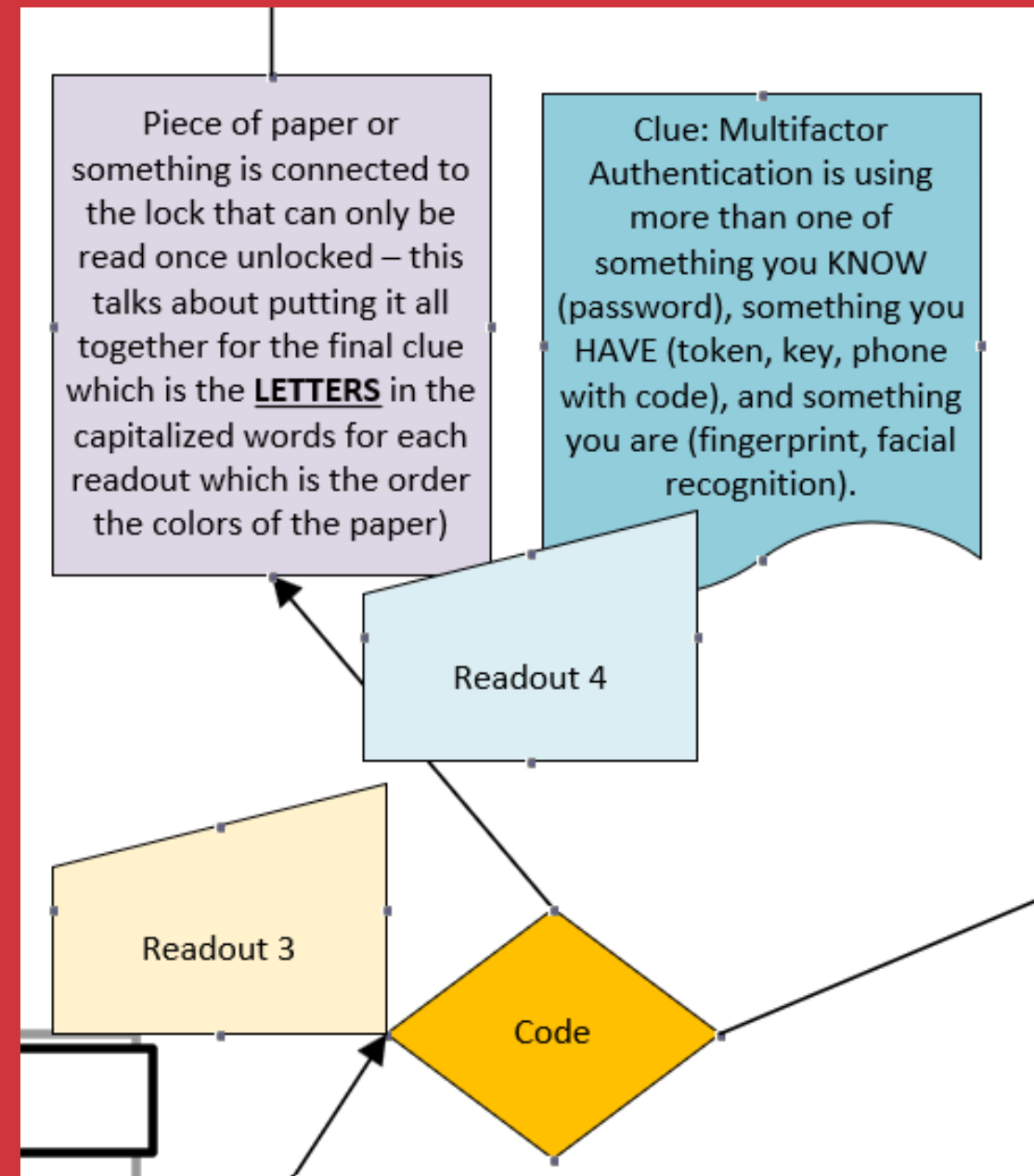
Lock 3 solved!

**MFA: Multi-Fun
Adventure**

Getting in this bag
requires two things...
Just solved one!



One factor down (something you know)!



Moves us into Encryption

FIRST:

Not only do these four things "Secure our World," they also can keep you (and your data) _ _ _ _

THEN: (On other side)

Using a Scytale Cipher (transposition cipher), can you find the corresponding encryption _ _ _ _ ?

Four ways to Secure our World (and data)

Readout for these four items when used with the overlay spells "SAFE" - Use with scytale cipher to get the password to get to the key

Strong passwords

multi-factor
Authentication

update soFware

rEport phishing

What's the word?

Wow! Looking back, these four things can “Secure our World.”

They also keep the Fed's data (and yours)  •

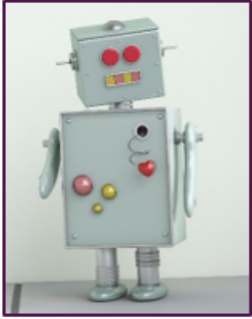
We all need to shut down daily, and that includes your work computer! When this happens, then the experts can **UPDATE SOFTWARE** behind the scenes! Up-to-date software is a great way to keep your devices secure (at home too).

STRONG PASSWORDS are so important for securing our data and access, but what if we could make authentication even stronger...

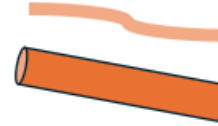
MULTIFACTOR AUTHENTICATION is using two or more different factors to authenticate that you are who you say you are.

Getting a suspicious email is bad enough and not clicking on a link is the right call. However, if you don't **REPORT PHISHING**, then we lose the ability to stop others from clicking it! When you report, it allows the experts the chance to protect others from clicking. Be the hero and report phishing!

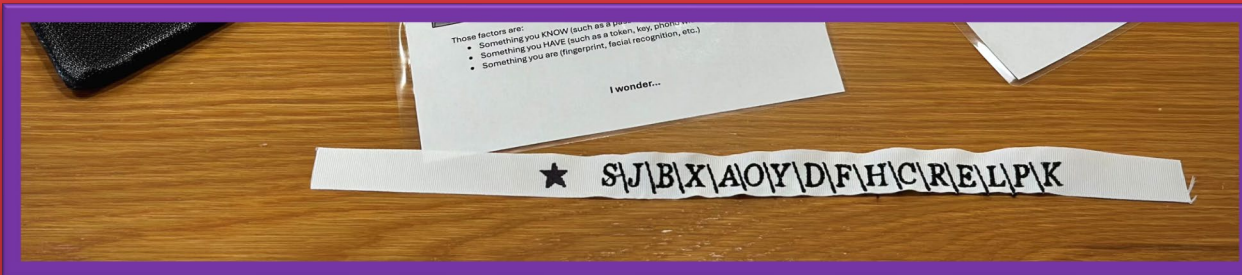
SAFE is the word, but not the password...



Have I told you about encryption? Encryption is the process that transforms readable data into an unreadable format. Ciphers can be used for encryption of some text. It's easier to WRAP your head around than you might think!



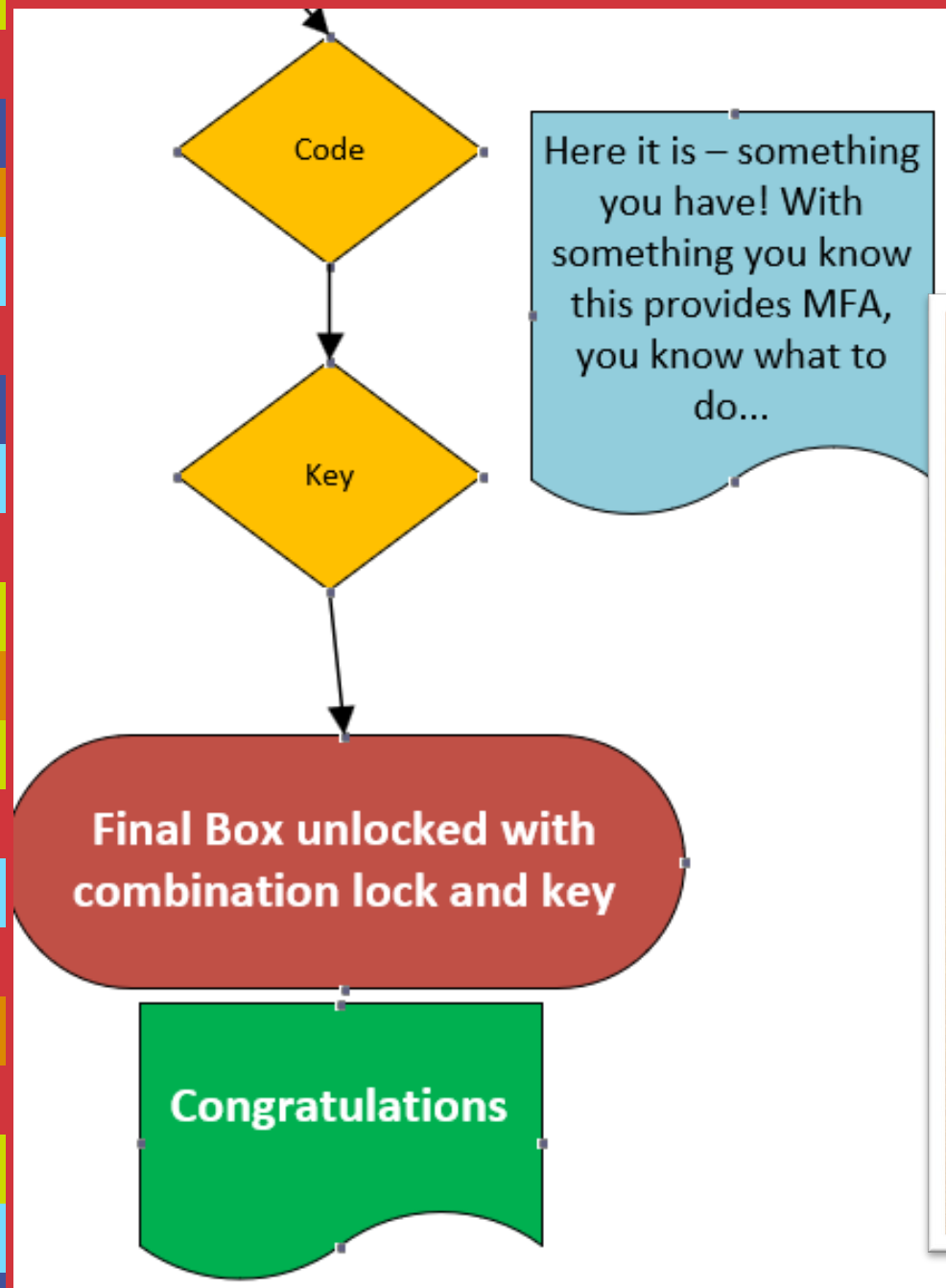
(First look at box)





**Second factor
coming up**





Escape Success!





Unlocked Wisdom

What we learned so
you don't have to.

- Build assembly instructions
- Do trial runs
- Get your leaders to play
- Get you knowledge champions involved
- Laminate what you can
- 4 to 6 people is the sweet spot to play
- Follow-up with an email with key concepts
- Anything you want them to read should be **BEFORE** the puzzles
- Make it a friendly competition!

Final thoughts from our Team

”Loved seeing the teamwork come out from not only the participants but from our team that built it as well.”
- Carolyn

“With one idea, we brought fun to the workplace, while changing the perception of InfoSec.”
-Brian

“Embrace the fun! If you are having fun creating it, chances are that your audience will enjoy participating.”
- Joanna

”There are many different ways to solve a problem.”
- Lori

”It was nice to create something that brought fun and excitement to an environment and topic that may seem stale and formulaic.”
- Brendan

Changing culture can be difficult, but taking a step to change culture doesn't have to be.



Enjoy your Escape!

Thank you

Joanna Behan

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